

Community Disaster Response Network (CDRN)

Project Overview

A real-time Community Disaster Response Network that connects citizens, volunteers, and authorities to manage disasters efficiently through incident reporting, alerts, and coordination.

Features

- Citizen incident reporting (Flood, Fire, Accident etc.)
- SOS emergency request system
- Authority dashboard for monitoring incidents
- Real-time data storage using Spring Boot REST APIs

Tech Stack

- Backend: Spring Boot
- Database: H2/MySQL
- Frontend: HTML, JavaScript
- Tools: Maven, REST APIs

Modules

1. Entity: Incident data model
2. Repository: Database operations
3. Service: Business logic
4. Controller: REST APIs

API Endpoints

POST /incident → Create incident
GET /incident → Fetch all incidents
GET /hello → Test API

How to Run

1. Clone repository
2. Open in VS Code
3. Run: mvn clean spring-boot:run

4. Open: <http://localhost:8080/hello>

Future Improvements

- Google Maps integration
- Real-time WebSocket alerts
- Volunteer management system
- AI-based disaster prediction